

Integrating Jenkins with Bitbucket and Jira: A Tutorial

This article gives an overview of Jenkins, Bitbucket and Jira. It also includes tutorials on how to install Jenkins and integrate it with Bitbucket and Jira.



Jenkins is a self-contained, open source automation server that can be used to automate all sorts of tasks related to building, testing and delivering or deploying software. As a prerequisite to following the various steps suggested in this article, you would need a public server for installing Jenkins.

Installing Jenkins

Jenkins can be installed through native system packages, Docker, or can even run standalone in any machine with a Java Runtime Environment (JRE) installed. As for now, we are installing Jenkins on a bare Ubuntu system.

The best practice is that whenever you deploy a server, you do an update and an upgrade with the following commands:

```
sudo apt-get update
```

```
sudo apt-get upgrade
```

Prerequisites

Since Jenkins is a Java application, the first step is to install Java. For that we need to configure the ppa (Personal Package Archive) repository for Java 8 OpenJDK package. Update the package index and install the Java 8 OpenJDK package with the following commands:

```
sudo add-apt-repository
ppa:webupd8team/java
sudo apt-get update
sudo apt-get install openjdk-8-jdk
```

Installation

Import the GPG keys of the Jenkins repository using the following `wget` command and after that configure the repository for Jenkins. Update the

package index and install the Jenkins package with the following commands:

```
wget -q -O - https://pkg.jenkins.io/
debian-stable/jenkins.io.key | sudo
apt-key add -
sudo apt-add-repository "deb https://
pkg.jenkins.io/debian-stable binary/"
sudo apt update
sudo apt install jenkins
```

Enabling and starting Jenkins service

After installing Jenkins, you need to enable Jenkins service using the `systemctl` command, so that on every reboot, Jenkins is automatically started. You can see the current status of the Jenkins service using the `systemctl status jenkins` command.

```
sudo systemctl enable jenkins
```